<table>
<tr><td colspan="3">Course Code: CS2226
Name of the Programme: B. Tech (H)
Name of the Course: Machine Learning for Data science
Semester: IV</td></tr>
<tr><td>Course credit</td><td>No. of hours per week (L + T + P)</td><td>Total no. of Teaching hours</td></tr>
<tr><td>3</td><td>(1:1:2)</td><td>60</td></tr>
<tr><td colspan="3">Course Objectives:</td></tr>
<tr><td colspan="3">a) Introduce the fundamental principles of neural networks and their role in the machine learning and data science ecosystem.</td></tr>
<tr><td colspan="3">b) Provide hands-on experience in building and training convolutional neural networks and sequences learning model using various modern architectures and techniques.</td></tr>
<tr><td colspan="3">c) Explore unsupervised learning approaches and generative models such as autoencoders and GANs for data representation and synthesis.</td></tr>
<tr><td colspan="3">d) Enable students to understand and apply model compression, optimization, and deployment strategies on edge devices and cloud platforms</td></tr>
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Course outline

Syllabus:

Module – 1 Machine learning Foundations for Data Science Applications	**No. of hours**
	3
Machine learning and deep learning in solving complex data science problems. Types of Machine Learning, Overfitting and Underfitting, Understand the architecture of shallow and deep neural networks, activation functions, and training mechanisms through forward and backward propagation. Explore common loss functions like Cross-Entropy and MSE, and various optimization techniques. Examine weight initialization strategies essential for robust model convergence in real-world datasets.	

Module – 2 Deep Learning for Visual Data Analytics	**No. of hours**
	3
Analyse image data using Convolutional Neural Networks (CNNs), a key tool in computer vision for data science tasks such as classification and object detection. Study CNN layers, pooling, and activation strategies. Delve into advanced convolution	

techniques—Depthwise Separable, Dilated, and Pointwise convolutions—and regularization methods like Batch Normalization and Dropout. Practice using pretrained models (e.g., VGGNet, ResNet, EfficientNet, DenseNet, Mobile net) and apply transfer learning, fine-tuning, and feature extraction

Module – 3 Sequential Data Analysis	**No. of hours**
	3
Modelling time-dependent or sequential data, such as stock prices, user behaviour, and natural language. Fundamentals of recurrent neural networks (RNNs)and limitations. LSTM and GRU architectures for improved sequence retention and learn their real-world applications in forecasting and NLP. Transformer architecture for data science applications.	

Module – 4 Unsupervised Learning for Feature Extraction and Data Generation	**No. of hours**
	3
Unsupervised learning techniques used in dimensionality reduction, and synthetic data generation for Data science applications. Unsupervised Learning and Generative Modelling with Autoencoders and GANs: Autoencoder (Types: Deep, CNN-based), Training Autoencoders, Variational Autoencoders (VAE), Introduction to GAN, GAN Architecture (Generator, Discriminator)	

Module – 5 Model Optimization and Deployment in the Data Science Pipeline	**No. of hours**
	3
Model compression techniques: Pruning, Quantization, Knowledge Distillation, ONNX and model conversion formats for deployment, Deep learning deployment on edge: NVIDIA TensorRT, OpenVINO, TFLite, Containerization and deployment: Docker, Flask/FastAPI, Streamlit, Deployment on cloud platforms: AWS SageMaker, GCP AI Platform, Azure ML.	

Course outcomes
a) Apply basic concepts of neural networks and deep learning in solving real time problems
b) Build deep learning models using convolutional and sequence learning techniques to process image, time-series, and text data.
c) Develop unsupervised learning models for Image Denoising, Image generation, anomaly detection.

d) Practices techniques for optimizing and deploying machine learning models in real-world environments.

Text Books	
1	Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. Deep learning. MIT Press, 2016, ISBN: 9780262035613.
2	Zhang, Aston, et al. "Dive into deep learning." Cambridge University Press, 2023, ISBN-13: 9781009389433.

Reference books	
1	Michael Nielsen. Neural Networks and Deep Learning. 2019. Online book.
2	Aurélien Géron. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. Shroff/O'Reilly. Third edition. ISBN-10: 1492032646
3	CS6910: Deep Learning, IIT Madras.
4	CS231n: Deep Learning for Computer Vision, Stanford University.
5	CS 4263/5483: Deep Learning, University of Texas at San Antonio
6	(6.S191) Introduction to Deep Learning, MIT

Tutorials	**[15 Hours]**
1	Linear Algebra for Neural Networks
2	Calculus for Deep Learning – Derivatives and Backpropagation
3	Optimization Techniques in Deep Learning
4	Mathematical Foundations of CNNs
5	Mathematics of Recurrent Networks and LSTMs
6	Generative Models and Latent Space Mathematics

Lab Programs /Tutorials	**[30 Hours]**
1	Building and Training a Basic Neural Network for Binary Classification
2	Implementing Forward and Backward Propagation from Scratch for Deep neural network
3	Designing and Training a Simple Convolutional Neural Network
4	Comparative Study of Modern CNN Architectures Using Transfer Learning
5	Developing Sequence Models Using RNN
6	Developing Sequence Models Using LSTM, and GRU
7	Building Deep and CNN based Autoencoders
8	Building Variational Autoencoders
9	Generating Synthetic Images Using GANs
10	Model compressions and deployment